

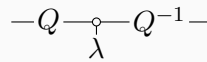
## Stress test: approach A on the book's hard diagram families

Old code is verbatim from the chapters. Line counts include `begin/end`.

### S1. Eigendecomposition (1.1) — copy dot with hanging $\lambda$

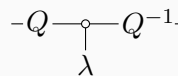
#### OLD (11 lines)

```
\begin{tikzpicture}[baseline=(Q.base), inner
  sep=1pt]
\node (Q) {$Q$};
\draw (Q.west) -- ++(-.3, 0);
\node[right=1em of Q, copydot] (dot) {};
\path (Q) edge (dot);
\node[below=.3em of dot] (e) {$\lambda$};
\path (dot) edge (e);
\node[right=1em of dot] (Qi) {$Q^{-1}$};
\path (dot) edge (Qi);
\draw (Qi.east) -- ++(.3, 0);
\end{tikzpicture}
```



#### NEW (7 lines)

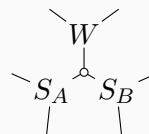
```
\begin{tikzpicture}[tn, baseline=(Q.base),
  every leg label/.style={}]
\node[legs={180}] (Q) {$Q$};
\node[right=1em of Q, copydot,
  legs={270/\lambda}] (dot) {};
\node[right=1em of dot, legs={0}] (Qi) {$Q^{-1}$};
\draw (Q) -- (dot) -- (Qi);
\end{tikzpicture}
```



### S2. Strassen star (decompositions) — hyperedge + 6 dangling legs

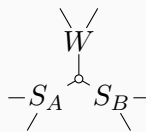
#### OLD (19 lines)

```
\begin{tikzpicture}[inner sep=1pt, baseline=.0
  em]
\node[copydot] (m) at (0,0) {};
\node (W) at (0,.5) {$W$};
\node (SB) at (-30:.5) {$S_B$};
\node (SA) at (-150:.5) {$S_A$};
\node (w1) at (60:1) {};
\node (w2) at (120:1) {};
\node (sa1) at (-120:1) {};
\node (sa2) at (-180:1) {};
\node (sb1) at (-0:1) {};
\node (sb2) at (-60:1) {};
\draw (m) -- (W);
\draw (m) -- (SA);
\draw (m) -- (SB);
\draw (W) -- (w1) {};
\draw (W) -- (w2) {};
\draw (SB) -- (sb1) {};
\draw (SB) -- (sb2) {};
\draw (SA) -- (sa1) {};
\draw (SA) -- (sa2) {};
\end{tikzpicture}
```



### NEW (8 lines)

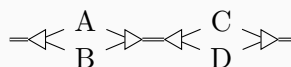
```
\begin{tikzpicture}[tn, baseline=0em, leg
length=.5]
\node[copydot] (m) at (0,0) {};
\node[legs={60,120}] (W) at (0,.5) {
$W$};
\node[legs={-120,180}] (SA) at (-150:.5) {
$S_A$};
\node[legs={0,-60}] (SB) at (-30:.5) {
$S_B$};
\draw (m) -- (W) (m) -- (SA) (m) -- (SB);
\end{tikzpicture}
```



## S3. Kronecker sandwich (511) LHS — where A does NOT win

### OLD (16 lines)

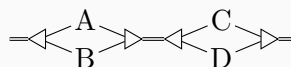
```
\begin{tikzpicture}[baseline=-.25em]
\node[triangle, rotate=180] (F0) at (0,0)
{};
\node (A) at (.6,.25) {A};
\node (B) at (.6,-.25) {B};
\node[triangle] (F1) at (1.2,0) {};
\draw[double] (F0) -- ++(-.4,0);
\draw (F0) -- (A) -- (F1);
\draw (F0) -- (B) -- (F1);
\node[triangle, rotate=180] (F2) at (1.8,0)
{};
\node (C) at (2.4,.25) {C};
\node (D) at (2.4,-.25) {D};
\node[triangle] (F3) at (3,0) {};
\draw (F2) -- (C) -- (F3);
\draw (F2) -- (D) -- (F3);
\draw[double] (F1) -- (F2);
\draw[double] (F3) -- ++(.4,0);
\end{tikzpicture}
```



Approach A alone: essentially the same 16 lines (legs only replace the two outer double stubs — the triangles and fan wiring remain hand-placed). The saving comes from approach D's molecule:

### NEW via D (7 lines)

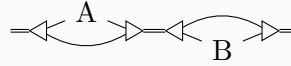
```
\begin{tikzpicture}[tn]
\pic (k1) {kron={A}{B}};
\pic (k2) at (1.8,0) {kron={C}{D}};
\draw[double] (k1-in) -- ++(-.4,0);
\draw[double] (k1-out) -- (k2-in);
\draw[double] (k2-out) -- ++(.4,0);
\end{tikzpicture}
```



## S4. (511b) — bent identity wires: honest tie

OLD (17 lines) — A offers nothing here

```
\begin{tikzpicture}[baseline=-.25em]
  \node[triangle, rotate=180] (F0) at (0,0) {};
  \draw[double] (F0) -- ++(-.4,0);
  \node (A) at (.6,.25) {A};
  \node[triangle] (F1) at (1.2,0) {};
  \draw (F0) -- (A) -- (F1);
  \draw (F0) edge[bend right] (F1);
  \node[triangle, rotate=180] (F2) at (1.8,0) {};
  \node (D) at (2.4,-.25) {B};
  \node[triangle] (F3) at (3,0) {};
  \draw (F2) edge[bend left] (F3);
  \draw (F2) -- (D) -- (F3);
  \draw[double] (F1) -- (F2);
  \draw[double] (F3) -- ++(.4,0);
\end{tikzpicture}
```



Bends between named nodes are already minimal in plain TikZ; a library can only add a `kron pic` variant with an identity wire. Verdict: tie.

## S5. Moment tensor $M_4$ — existing macros already win

OLD (8 lines, uses “`vecmatvec`”)

```
$M_4 =
\renewcommand*\arraystretch{1.3}
\begin{bmatrix}
  \vecmatvec{1em}{\tilde x}{}{} \backslash \backslash
  \vecmatvec{1em}{\tilde x}{}{} \backslash \backslash
  \vecmatvec{1em}{\tilde x}{}{} \backslash \backslash
  \vecmatvec{1em}{\tilde x}{}{} \backslash \backslash
\end{bmatrix}$
```

$$M_4 = \begin{bmatrix} \tilde{x} \\ \tilde{x} \\ \tilde{x} \\ \tilde{x} \end{bmatrix}$$

The book’s `\vecmatvec`, `\trace`, `\detstack` macros are already the right abstraction for their patterns. Approach A complements them; it should not replace them.

## S6. Order-4 tensor with labeled legs — A’s best case

OLD style (9 lines)

```
\begin{tikzpicture}[baseline=-.25em, inner sep=1pt]
  \node (T) {$T$};
  \draw (T) -- ++(-.4,.3)
    node[pos=1.3, font=\tiny] {i};
  \draw (T) -- ++(.4,.3)
    node[pos=1.3, font=\tiny] {j};
  \draw (T) -- ++(-.4,-.3)
    node[pos=1.3, font=\tiny] {k};
  \draw (T) -- ++(.4,-.3)
    node[pos=1.3, font=\tiny] {l};
\end{tikzpicture}
```



NEW (3 lines)

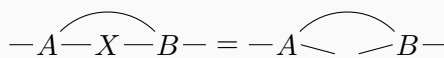
```
\begin{tikzpicture}[tn]
  \node[legs={143/i, 37/j, 217/k, -37/l}] (T) {
    $T$};
\end{tikzpicture}
```



## S7. Delete-the-node figure — both sides, verbatim

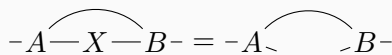
### OLD (10 + 9 lines, verbatim)

```
\begin{tikzpicture}[baseline=-.25em, inner sep
=1pt]
  \node (A) at (0,0) {$A$};
  \node (X) at (.8,0) {$X$};
  \node (B) at (1.6,0) {$B$};
  \draw (A) -- (X);
  \draw (X) -- (B);
  \draw (A) edge[bend left=45] (B);
  \draw (A) -- ++(-.5,0);
  \draw (B) -- ++(.5,0);
\end{tikzpicture}
$=$
\begin{tikzpicture}[baseline=-.25em, inner sep
=1pt]
  \node (A) at (0,0) {$A$};
  \node (B) at (1.6,0) {$B$};
  \draw (A) edge[bend left=45] (B);
  \draw (A) -- ++(-.5,0);
  \draw (B) -- ++(.5,0);
  \draw (A.east) -- ++(.45,-.12);
  \draw (B.west) -- ++(-.45,-.12);
\end{tikzpicture}
```



### NEW (7 + 6 lines)

```
\begin{tikzpicture}[tn]
  \node[legs={180}] (A) {$A$};
  \node[right=1em of A] (X) {$X$};
  \node[right=1em of X, legs={0}] (B) {$B$};
  \draw (A) -- (X) -- (B);
  \draw (A) edge[bend left=45] (B);
\end{tikzpicture}
$=$
\begin{tikzpicture}[tn]
  \node[legs={180, -20}] (A) {$A$};
  \node[right=2.9em of A, legs={0, 200}] (B) {
    $B$};
  \draw (A) edge[bend left=45] (B);
\end{tikzpicture}
```



Modest win: the deleted-node stubs are just more legs (`legs={180, -20}`).

## Verdict

| Family                                                                      | Old   | New | Winner                  |
|-----------------------------------------------------------------------------|-------|-----|-------------------------|
| Derivative rows (loops)                                                     | 8–12  | 5–7 | A + <code>\dloop</code> |
| Labeled-leg tensors, TT chains                                              | 6–10  | 2–4 | A (big)                 |
| Stars / hyperedges (Strassen)                                               | 19    | 8   | A (big)                 |
| Copy-dot chains (eigen, Hadamard)                                           | 11    | 7   | A (modest)              |
| Bundle sandwiches (Kronecker)                                               | 14–17 | 6–7 | D pics, not A           |
| Bent wires, twists, wraps                                                   | —     | —   | tie: plain TikZ         |
| <code>\vecmatvec</code> / <code>\trace</code> / <code>\detstack</code> rows | 1–8   | —   | existing macros         |

A is the right *foundation* (it never loses, wins wherever legs and labels dominate), D's pics handle the recurring molecules, and the existing macros stay. The one honest gap: nothing shortens bent-wire plumbing.